Heating Effect of an Electric Current

High resistance in a conductor produces heat.

Electrical energy is converted into heat energy

Conductors which are long, have small diameters and have a high resistivity produce the most heat.

Nichrome and tungsten metals have high resistances

Appliances which convert electrical energy to heat are:

electric heater	**(nichrome)**
light globe	**(tungsten)**
toaster	**(nichrome)**
electric kettle	

Magnetic Effect of an Electric Current

A conductor carrying an electric current produces a magnetic field.

When the conductor is wound on a soft iron core it is an electro magnet.

The conductor only acts as a magnet as long as the current is flowing.

Electro magnets have many varied uses: cranes, generators, motors